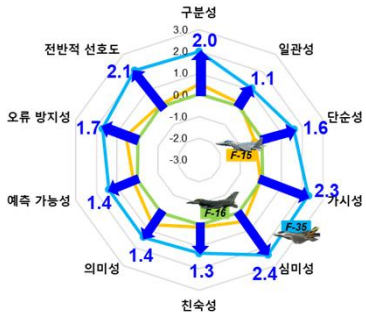
Vision &
Visual Display Design



A study of LAD/HUD PVI design methodology considering human factors (You et al., June 2016 ~ June 2017; sponsored by Korean Aerospace Industry)

Contents

Part I

- Visual Capabilities
- Hardcopy Text Design
- VDT Screen Design
- Graphics, Symbols & Codes

Part II

- Dynamic & Quantitative Displays
- Object Displays & Integration
- Qualitative Displays & Signals
- Representational & Advanced Displays
- Quiz

Learning Goals (1/2)

GOALS

- Explain the **primary uses of dynamic displays**, distinguishing among **quantitative, qualitative, check readings, and situation awareness**.
- Evaluate the task suitability of **Analog versus Digital quantitative displays**.
- Describe **Parallax Errors** in scale readings and apply **pointer design guidelines** to minimize reading errors.
- Explain the **Compatibility of Proximity Principle** by analyzing the relationship between **mental proximity and physical display proximity**.
- Describe **Emergent Features**, such as shape or area, and explain how they facilitate simultaneous parallel processing in object displays.
- Evaluate the performance **trade-offs of Object Displays**, comparing their advantages for system state integration against their disadvantages for focused attention and precise reading tasks.

- Explain how **color and shape coding in qualitative displays** reduce cognitive load by dividing continuous scales into distinct operational zones.

Learning Goals (2/2)

GOALS

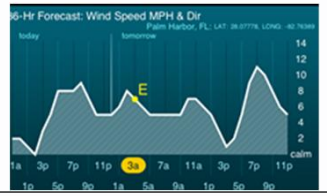
- Apply **Gestalt principles of similarity and continuity** to panel monitoring by aligning pointers to enable the pre-attentive pop-out effect in **check reading**.
- Evaluate **signal and warning light configurations** based on human factors guidelines, specifying appropriate **contrast ratios, flash rates, and visual angles for critical emergencies**.
- Define the **three levels of Situation Awareness**, including **Perception, Comprehension, and Projection**, in the context of monitoring complex dynamic systems.
- Evaluate **Representational Display** formats using **Roscoe's principles of pictorial realism, compatible motion, integration, and pursuit presentation**.
- Explain the **dark focus trap** associated with **collimated Head-Up Displays (HUDs)** and evaluate its resulting **perceptual distortions regarding object size and distance**.
- Recognize the **future trends in visual displays**.

Section 05

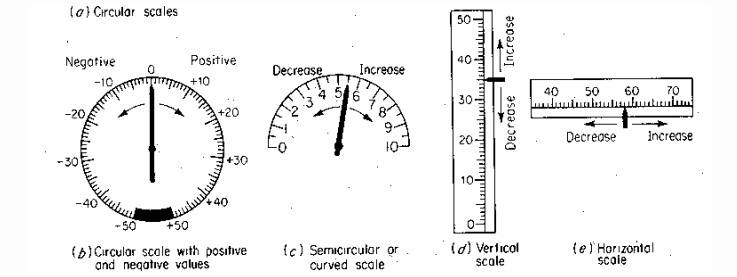
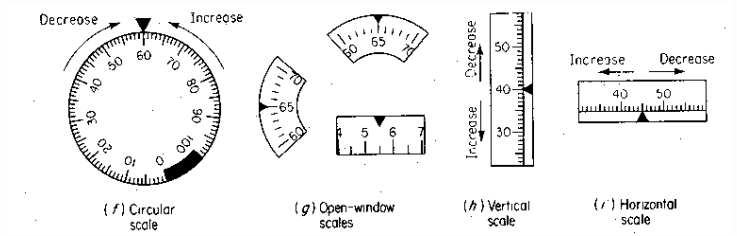
Dynamic Information & Quantitative Displays

Principles for designing displays that present changing information, trends, and precise numerical values

Primary Uses of Dynamic Displays

Usage	Description	Good Design
Quantitative readings	Used to read a precise numeric value or exact magnitude.	
Qualitative readings	Used to discern an approximate value, trend, rate of change, or change in direction	
Check readings	Used to quickly determine if parameters are within "normal" safety bounds	
Situation awareness	Used to **perceive the current status** of the system in time & space and **project its status** into the near future	

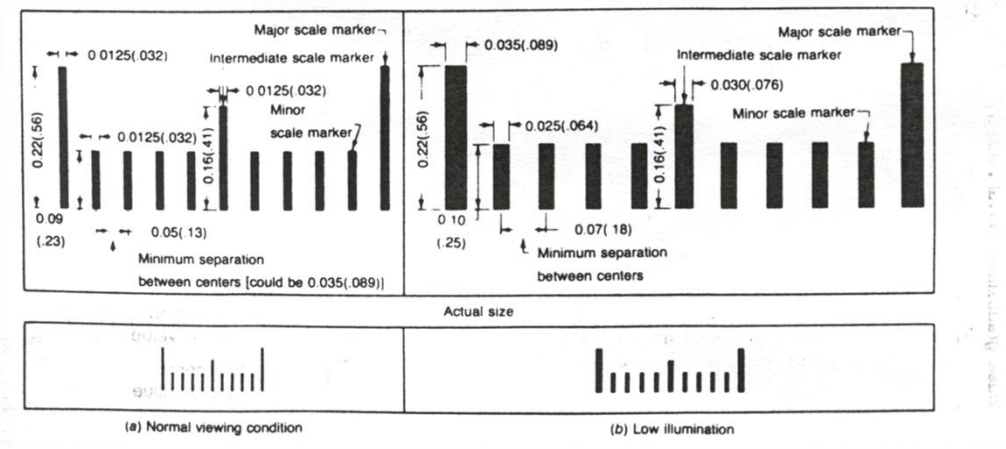
Types of Quantitative Displays

Use		Description	Good Design
Analog display	**Fixed scale** with moving pointer	• Scale is static; pointer moves to indicate value • Most common analog type • Avoids reversal errors • Better cue for zero/null position	
	Moving scale with fixed pointer	• Pointer is static; scale rotates behind it • Used in compasses/altimeters. • Prone to reversal errors • Harder to visualize range	
Digital display		• Direct numeric readout without interpretation of position. • Precise reading (no interpolation) • Poor for rate of change/trends	

Quantitative Display Designs

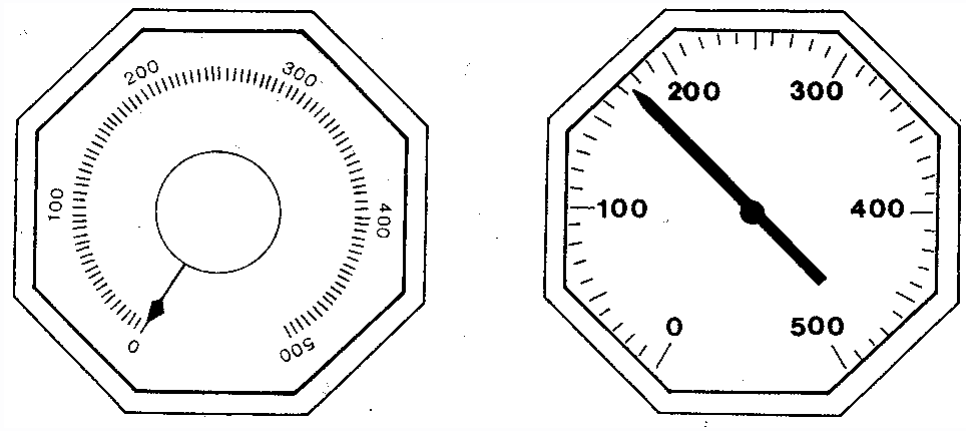
Design A

Design B



Which is preferred?

Design A



Design B

Analog vs. Digital Displays: When to Use Which?

Golden Rule: Match the display type to the TASK requirement!

Analog Display

- **Direction or RATE of change** is important
- Values change **frequently or rapidly**
- Quick **trend detection** or qualitative check is needed

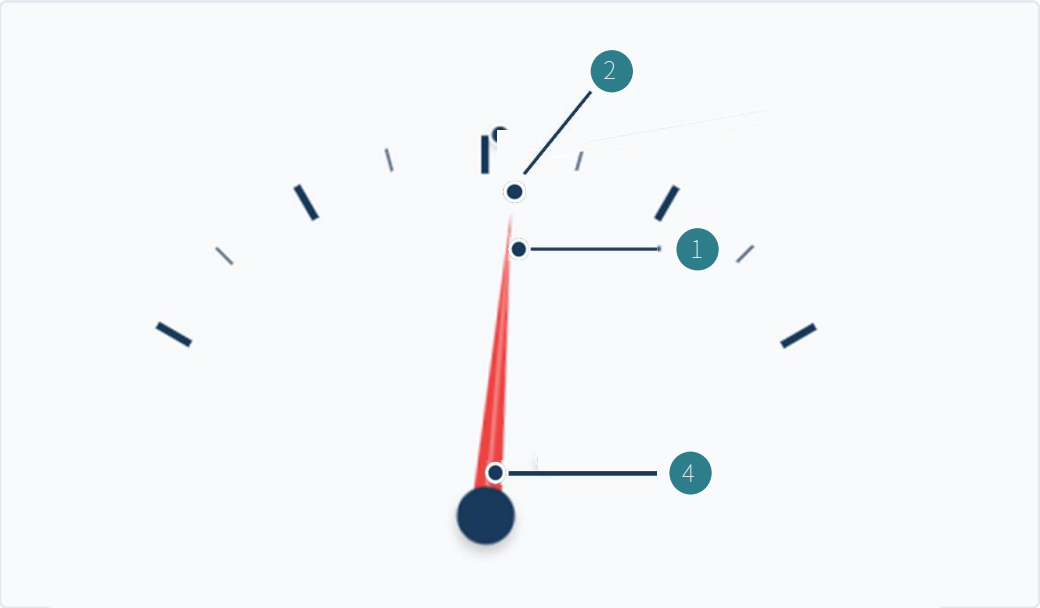
Example: Speedometer while driving

Digital Display

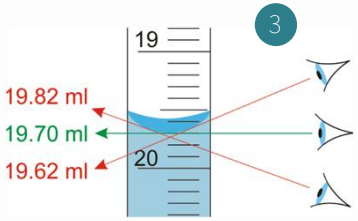
- **Precise numeric value** is required
- Values change **slowly** or remain stable enough to be read
- Task involves **recording exact numbers**

Example: Fuel gauge reading during refueling

Pointer Design Guidelines



Parallax: The displacement or difference in the apparent position of an object viewed along two different lines of sight.



1 Tip Angle (~20°)

Pointer tip should be tapered to a sharp angle of approximately 20° to allow for precise reading of values.

2 Scale Relationship

The tip should **meet**, but not overlap, the smallest scale markers. Overlapping obscures the marker; a gap increases estimation error.

3 Minimize Parallax

The pointer should be mounted as close to the scale surface as possible.
(Large gap + angle viewing = reading error)

4 Color & Visibility

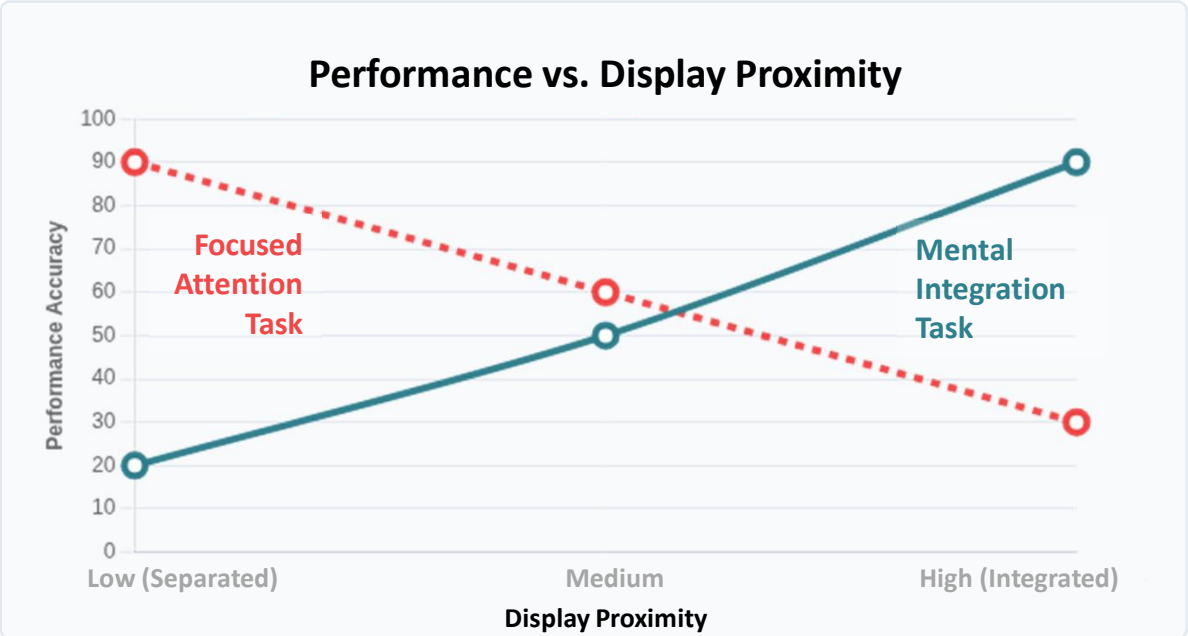
The pointer color should extend from the tip all the way to the center of rotation (axle) to maximize visual detection.

Section 06

Object Displays & Information Integration

Exploring the compatibility of proximity principle, emergent features, and task suitability for integrated display design

Compatibility of Proximity Principle



Mental Integration Task
Combining variables (A + B)
✓ **Close display proximity helps**

Focused Attention Task
Isolating variable (Just A)
× **Close display proximity clutters**

Design Goal: Match **display proximity** to the task's required **mental proximity**.

Mental Proximity

The extent to which a task requires the **cognitive integration of multiple information sources**.

Display Proximity

The extent to which multiple information sources are physically (in space) or perceptually grouped (by shared visual attributes like color or code).

The Conflict

High display proximity helps **a mental integration task** but hurts a focused attention task due to visual clutter and difficulty separating variables.

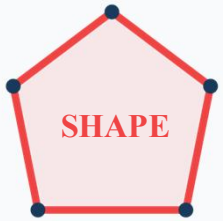
Emergent Features

1. Separate Parts

Two independent variables
(e.g., pressure & temp)

2. Rectangle

Emergent Feature:
AREA
New dimension appears



3. Polygon

Emergent Feature:
SHAPE
Pattern reveals system state

Definition

An **Emergent Feature** is a **perceptual property of the whole** that does **not exist in the individual parts**.

Perceptual Benefit

Allows operators to process multiple variables simultaneously **(parallel processing)** rather than sequentially.

Application: Object Displays

When variables are mapped to a geometric object (like a polygon), a **"normal" system state creates a symmetric shape**. An anomaly distorts the shape, making it instantly recognizable.

Object Displays: Pros & Cons

When They Help

Object Display Advantage

- **Integration Tasks:** Best when operator needs to combine multiple variables mentally.
- **Pattern Recognition:** Changes in system state create recognizable shape distortions.
- **Rapid State Detection:** Quickly answers "Is the system normal overall?"

Ideal Application

Power plant safety parameter monitoring

When They Hurt

Object Display Disadvantage

- **Focused Attention Tasks:** Poor for isolating a single variable from the whole.
- **Precise Reading:** Harder to extract exact numeric values compared to digital displays.
- **Specific Query:** Slows down answers to "What is the exact RPM value?"

Better Alternative Here

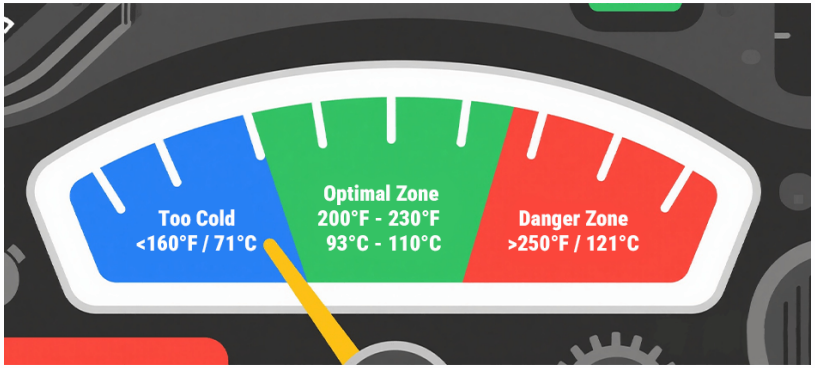
Separate digital counters for precise data logging

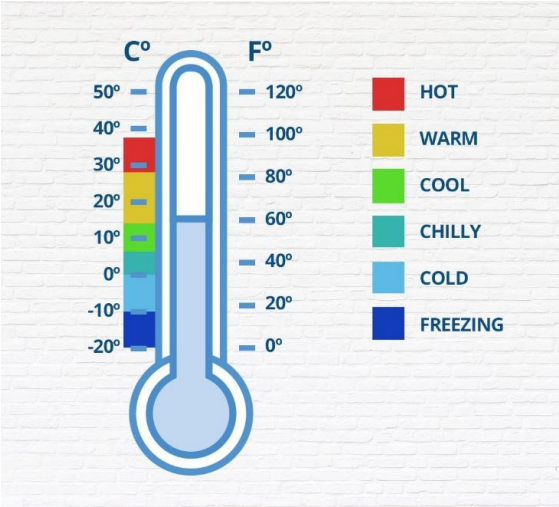
Section 07

Qualitative Displays, Check Reading & Signal Lights

Design principles for range-based indicators, pattern recognition in panel monitoring, and effective warning systems

Qualitative Displays





Qualitative Reading

Used when the user needs an **approximate value**, trend, or rate of change rather than a precise number (e.g., "Is it safe?" not "Is it 87.3°?").

Color Coding

Divides the scale into distinct zones: **Green** (Normal), **Yellow** (Caution), **Red** (Danger). Reduces cognitive processing time.

Shape Coding

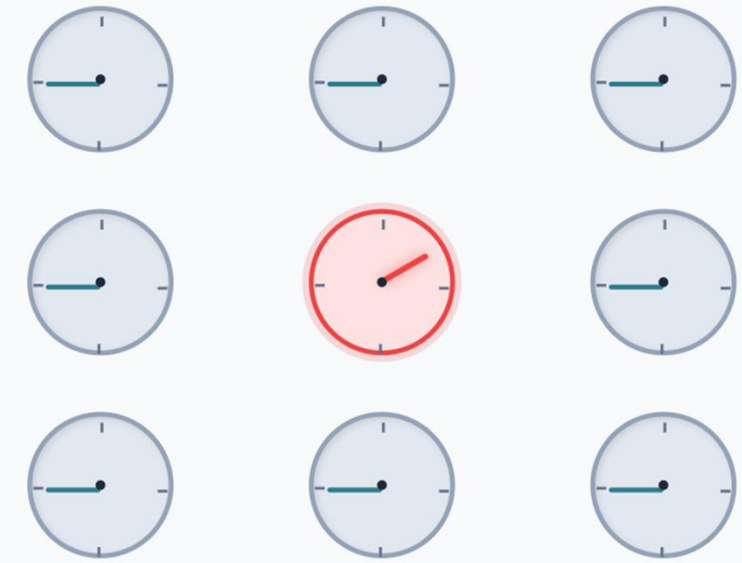
Uses specific shapes (e.g., icons, patterns) within zones to aid recognition, especially for color-deficient users or in low light.

Benefits

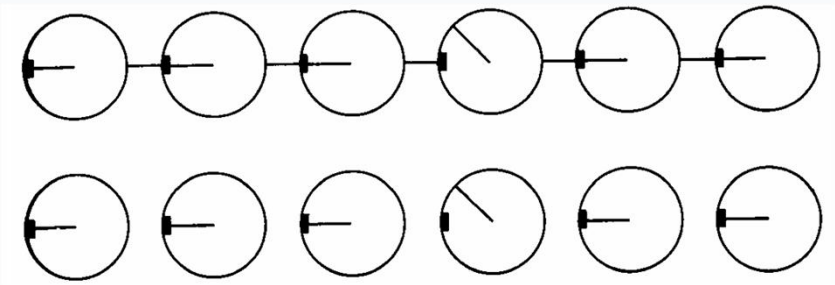
Faster perception, reduced cognitive load, and immediate recognition of system state (Safe vs. Unsafe).

Check Reading: Using Gestalt for Panel Monitoring

Visual Similarity



Visual Continuity



9 O'clock Alignment

Aligning all "normal" pointers to the 9 o'clock position (or 12 o'clock) creates a consistent horizontal line.

Gestalt Principles

Uses **Similarity** and **Continuity**. The brain perceives the aligned pointers as a single group or pattern.

Pop-out Effect

A single deviant pointer breaks the pattern and "pops out" pre-attentively, allowing instant detection of abnormal conditions.

Key Benefit: Operators can check 50+ dials in a single glance rather than reading each value individually.

Signal & Warning Lights

Optimal Warning Light

Red Signal Light (Danger)

Contrast Ratio	> 2:1
Visual Angle	≥ 1°

Steady vs. Flashing

Use **steady lights** for ongoing conditions. Reserve **flashing lights** for new or emergency conditions to attract immediate attention.

Color Code

Use **Red** for danger/warning. It is universally associated with hazard and captures attention effectively.

Flash Rate

Optimal rate is **3 to 10 Hz** (flashes per second) **with duration of at least 0.05s**. Avoid rates that induce seizures.

Minimum Size

Should subtend at least **1° of visual angle** to ensure high detectability across varied conditions.

Intensity

Luminance should be at least **2× brighter than the immediate background** (contrast ratio ≥ 2:1).

Location

Place within **30° of the operator's normal line of sight** for peripheral detection.

Max Flash Rates

Do not use more than **3 distinct flash rates** to code different levels of urgency (risk of confusion).

Section 08

Representational Displays & Advanced Technologies

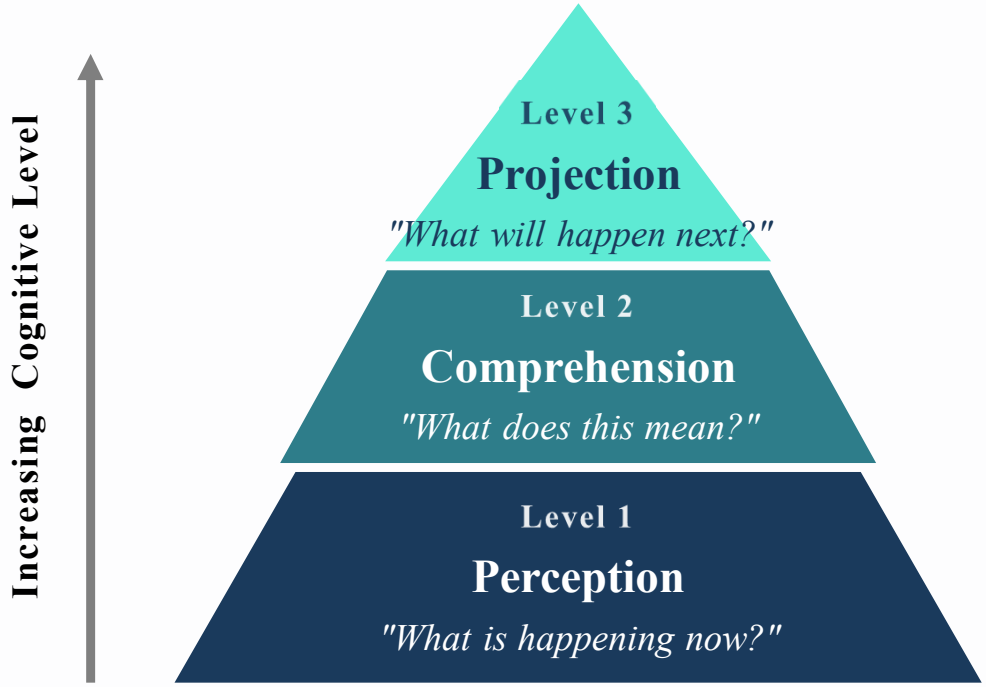
Exploring situation awareness, aircraft display principles, 3D vs 2D perspectives, and HUD challenges

Situation Awareness

Situational awareness (SA) is the **ability to perceive, understand, and anticipate1 what is happening in immediate future**. It involves recognizing critical elements, comprehending their meaning, and predicting future actions to **make informed, safe decisions, often under pressure**.

Critical Applications

- Air Traffic Control
- Cockpit Displays
- Emergency Medicine
- Military Operations



Level 3: Projection

Highest level. Forecasting future status of elements based on current understanding. Critical for proactive decision making.

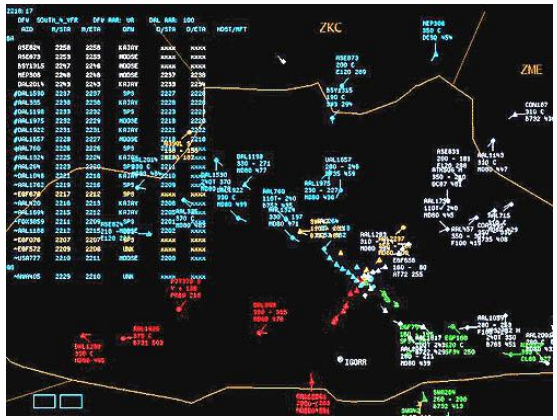
Level 2: Comprehension

Synthesizing disjointed Level 1 elements. Understanding significance of objects and events in relation to goals.

Level 1: Perception

Foundation level. Perceiving status, attributes, and dynamics of relevant elements in the environment.

Situation Awareness: Air Traffic Control



Situation Awareness: Cockpit Displays

Aircraft Bank Angle Displays

Outside-In Display

"Bird's Eye"

Moving Aircraft, Fixed Horizon

- **Horizon is Fixed** relative to the instrument frame
- **Aircraft symbol Moves** to show bank angle
- Bank Left → Indicator rotates LEFT ⬅
- Perspective: **Viewing the plane from the ground**

Inside-Out Display

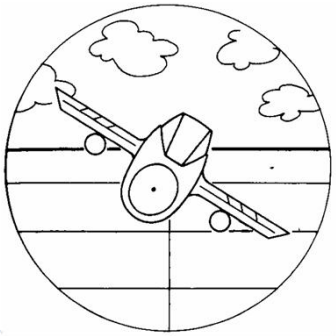
"Pilot's Eye"

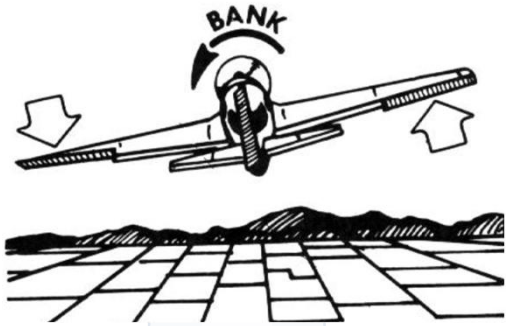
Fixed Aircraft, Moving Horizon

- **Aircraft symbol is Fixed** in the center
- **Horizon line Moves** to show bank angle
- Bank Left → Horizon rotates Right ➡
- Perspective: **Viewing out from the cockpit window**

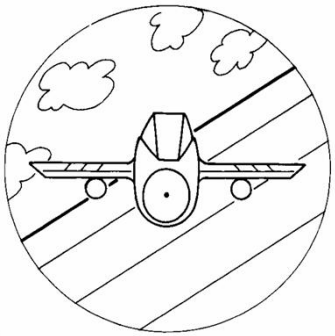
Bank Left

Moving Aircraft, Fixed Horizon





Fixed Aircraft, Moving Horizon



Roscoe's Principles for Aircraft Displays (1/2)

Roscoe et al. (1981) crystallized four **key principles to guide the optimal design of representational displays**, such as aircraft-position displays

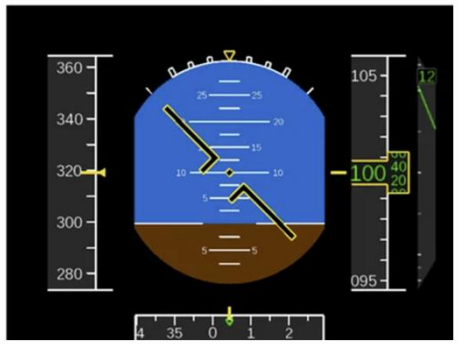
Principle of Pictorial Realism

The display should **present a spatial, analog representation of the real-world environment it represents**, maintaining correct depth cues and up/down, left/right frames of reference.

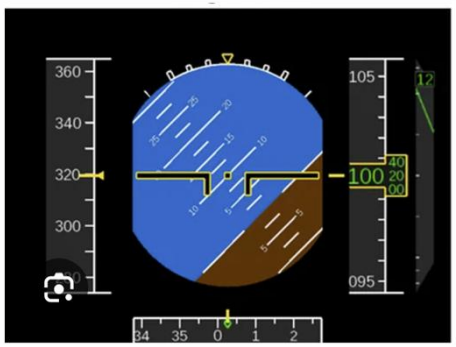
Principle of Compatible Motion

Display movement should be **congruent with the operator's frame of reference**. Supports the use of **moving-aircraft (outside-in) displays** with fixed horizons for pilots.

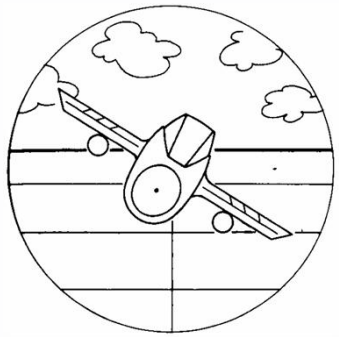
Moving Aircraft, Fixed Horizon



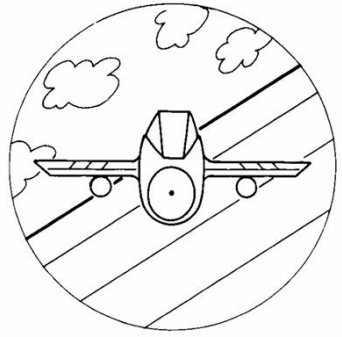
Fixed Aircraft, Moving Horizon



Moving Aircraft, Fixed Horizon



Fixed Aircraft, Moving Horizon



Roscoe's Principles for Aircraft Displays (2/2)

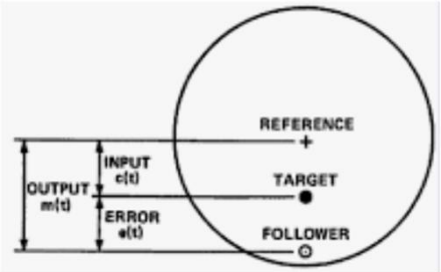
Principle of Integration

Related information should be **presented in a single integrated display** rather than scattered across separate instruments to reduce scanning time.

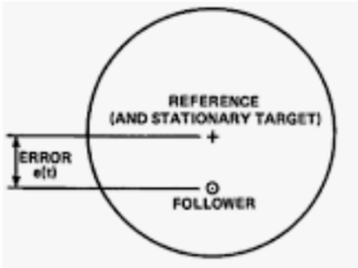
Principle of Pursuit Presentation

Preference for **pursuit displays (where both target and follower move)** over compensatory displays (where only error is shown) for better tracking performance.

Pursuit Display



Compensatory Display



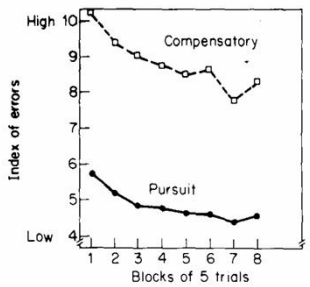


FIGURE 10-13.
Comparison of errors of subjects when using a compensatory versus a pursuit display in a tracking task. (*Source: Adapted from Briggs and Rockway, 1966, Fig. 2. Copyright © 1966 by the American Psychological Association. Reprinted by permission.*)

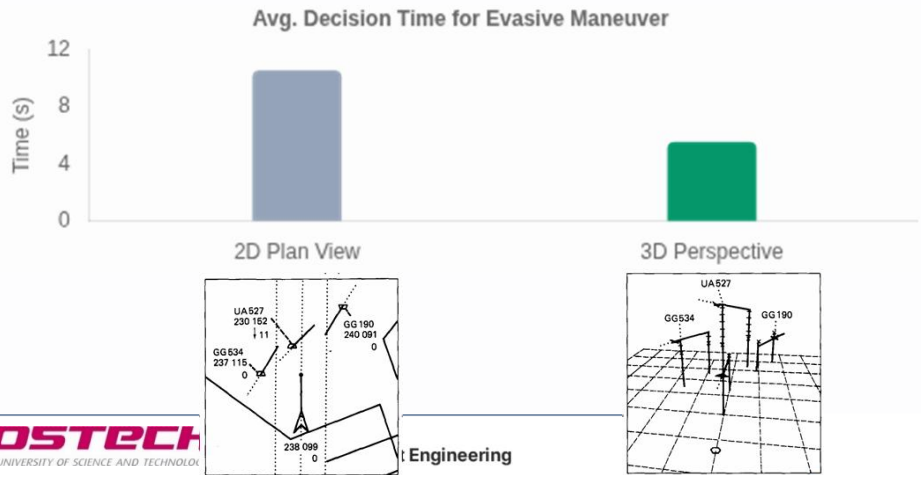
2D Plan vs. 3D Perspective View

3 ~ 6 sec

Faster Decision Time

using 3D perspective displays

> "Pilots initiated evasive maneuvers significantly faster with 3-D displays compared to 2-D plan view displays for most traffic scenarios."
>
> — **Ellis, McGreevy, & Hitchcock (1987)**



Why 3D Displays Perform Better

Natural Depth Cues

3D perspective displays utilize linear perspective, gradients, and occlusion, providing intuitive depth information that matches human visual experience in the real world.

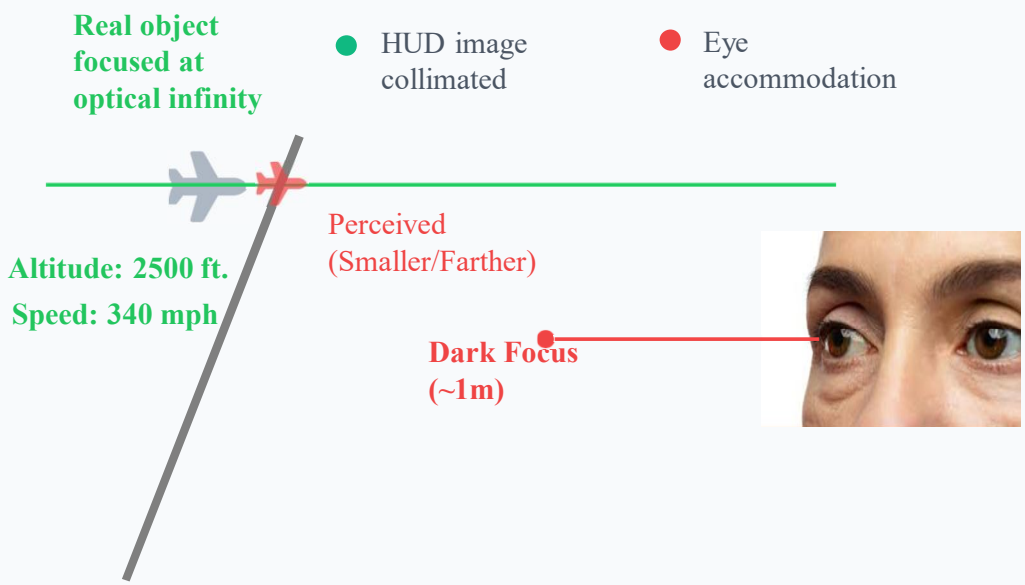
Limitations of 2D Plan View

Traditional 2D displays provide a flat, top-down view. Pilots must mentally integrate altitude and position data, increasing cognitive load and reaction time.

Important Exception

For **head-on traffic**, 3D displays were actually SLOWER due to ambiguity in determining the exact direction of the approaching aircraft.

HUD (Head-Up Display) & Dark Focus Problem



HUD Optical Design

HUD images are collimated (focused at optical infinity) so pilots theoretically don't need to re-accommodate from the outside world.

The Dark Focus Trap

Despite infinity focus, the eye tends to lapse into its resting state ("dark focus") at ~1 meter (arm's length) when viewing HUDs.

Perceptual Consequences

Accommodation at 1m causes distant objects to appear **smaller** and **farther away** than they actually are (micropsia).

⚠ Safety Risk

Pilots may misjudge distance to the ground or other aircraft, potentially delaying evasive maneuvers or flaring too late during landing.

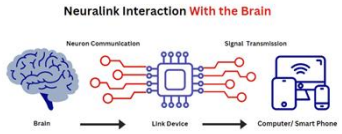
Display Design Decision Framework

What information does the user primarily need?

Precise numeric value?	Trend, rate, or direction?	Normal vs. Abnormal status?	Integrated system state?	Immediate Warning?
Digital Display Best for exact reading & recording	**Analog Display** Best for quick check of change/direction	**Qualitative / Check** Ranges, zones, & aligned dials	**Object Display** Polygons for pattern recognition	**Signal Light** Red flashing (≥1°) for emergency

Future Trends in Visual Displays

The Future: Moving from static presentation to adaptive and intelligent symbiosis.

Trend	Description	Example
Adaptive Displays	Interfaces that automatically adjust information density and format based on user workload and physiological state	
Eye-Tracking Integration	Gaze-contingent displays that render high detail only where the user looks, optimizing processing and control	
AI-Driven Optimization	Machine learning algorithms that predict user intent and dynamically reorganize the interface for efficiency.	
Mixed Reality Cockpits	Seamless blending of physical controls with virtual overlays to enhance situational awareness without isolation.	
Neural Interface Displays	Direct brain-computer interfaces (BCI) allowing intent-based control and information presentation.	

HF/E & Aerospace Sectors (KAI & KASA)

Aerospace Human Factors Engineering Workshop